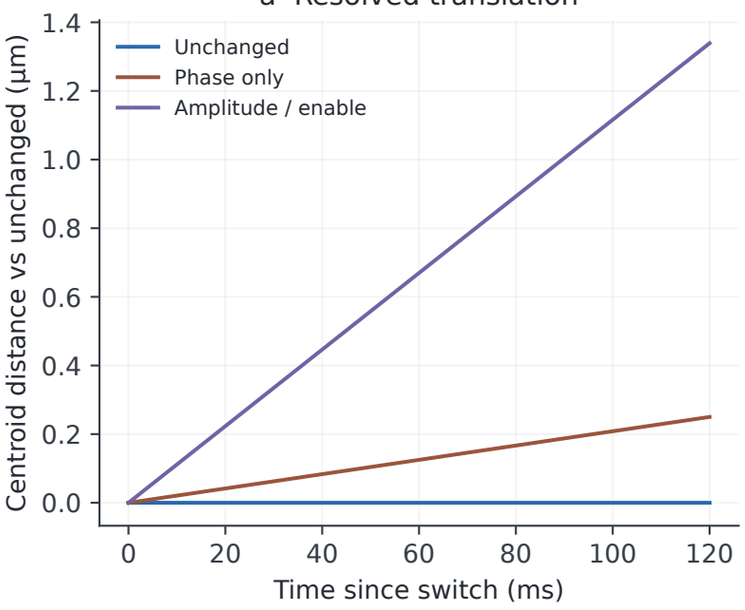
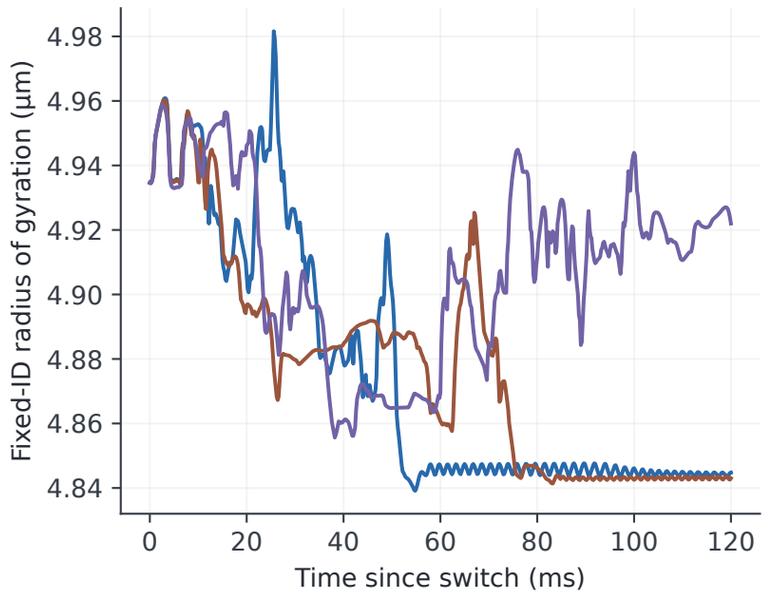
F3-3 | PZT control versus the same-state unchanged trajectory
Stable reduction · 120 ms continuation · fixed criteria and separately measured sensitivities

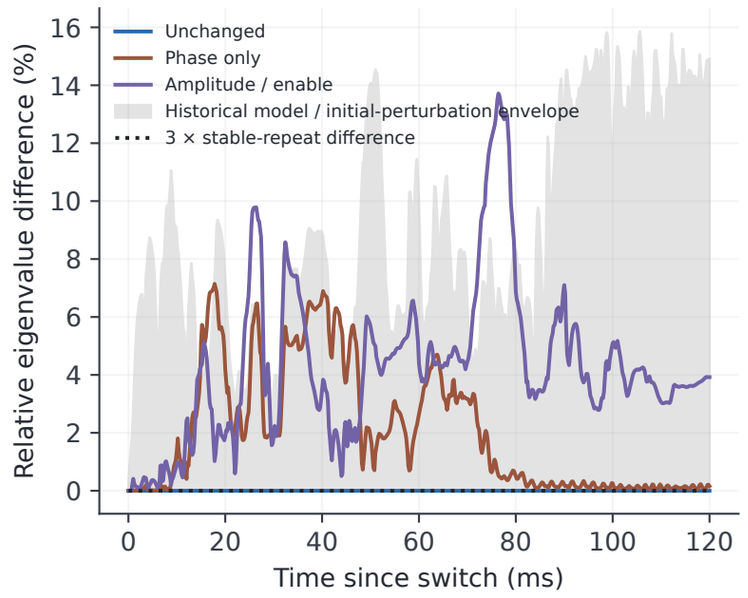
a Resolved translation



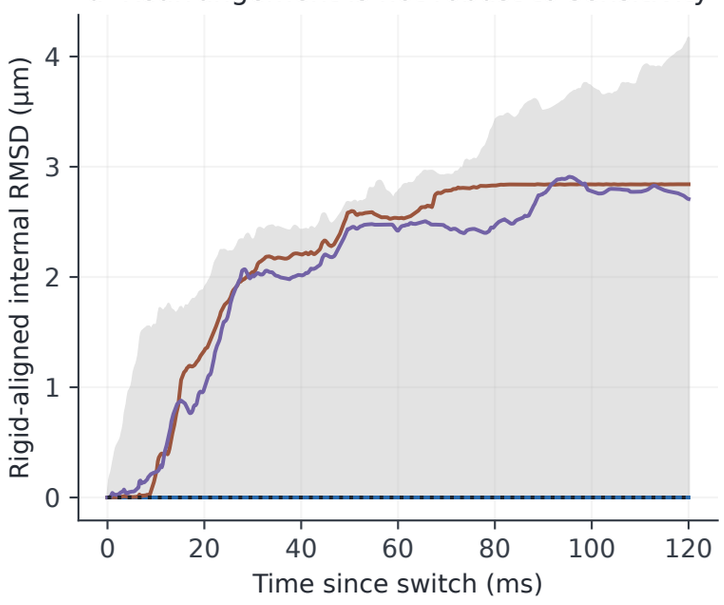
b Cluster size; stable repeats overlap



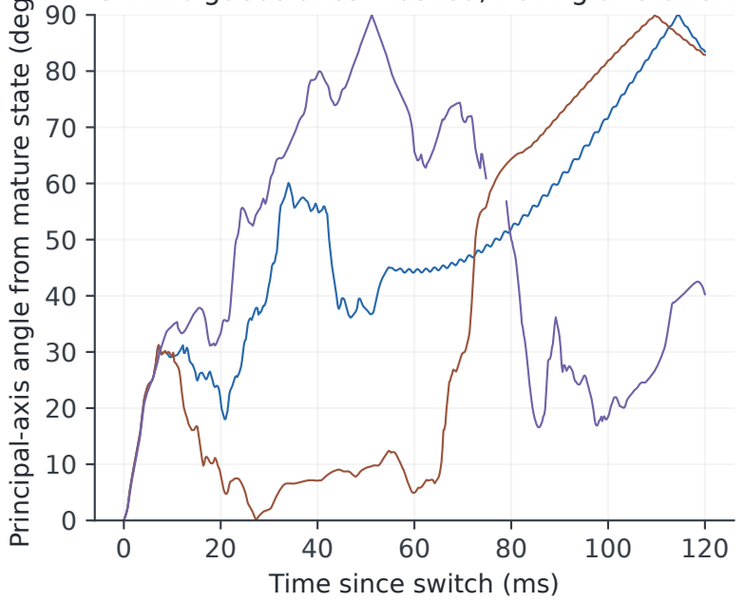
c Shape contrast and sensitivity envelope



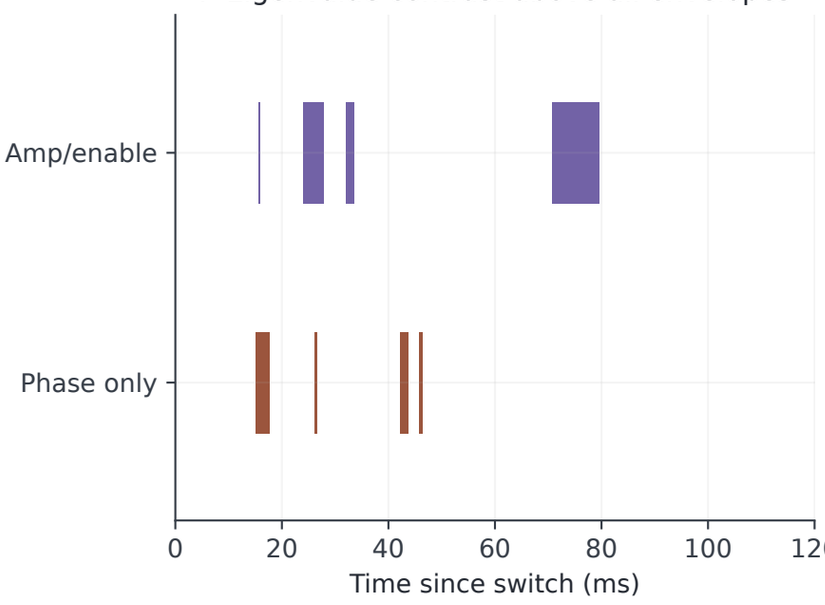
d Rearrangement is not robust to sensitivity



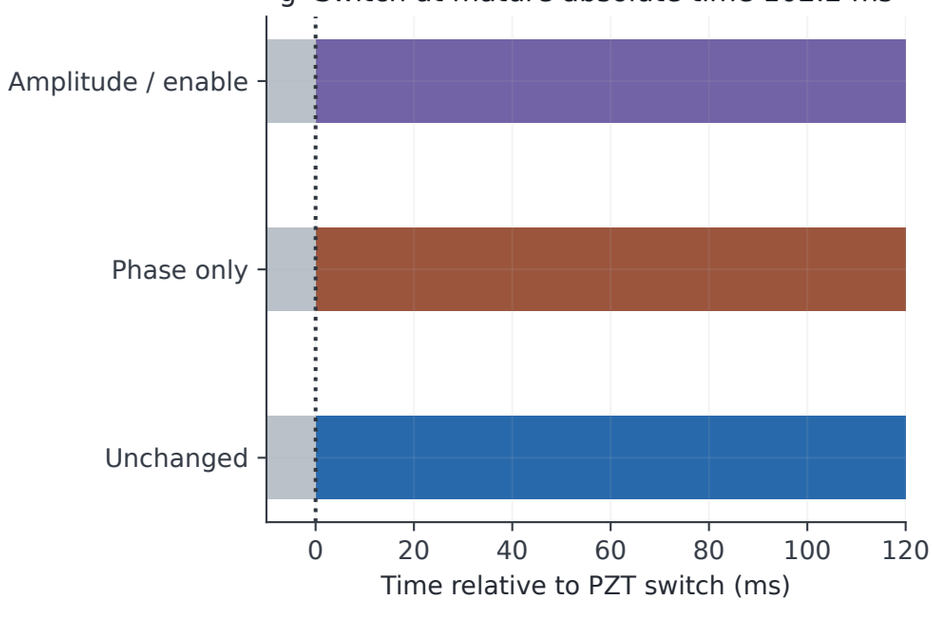
e Ambiguous axes masked; not rigid rotation



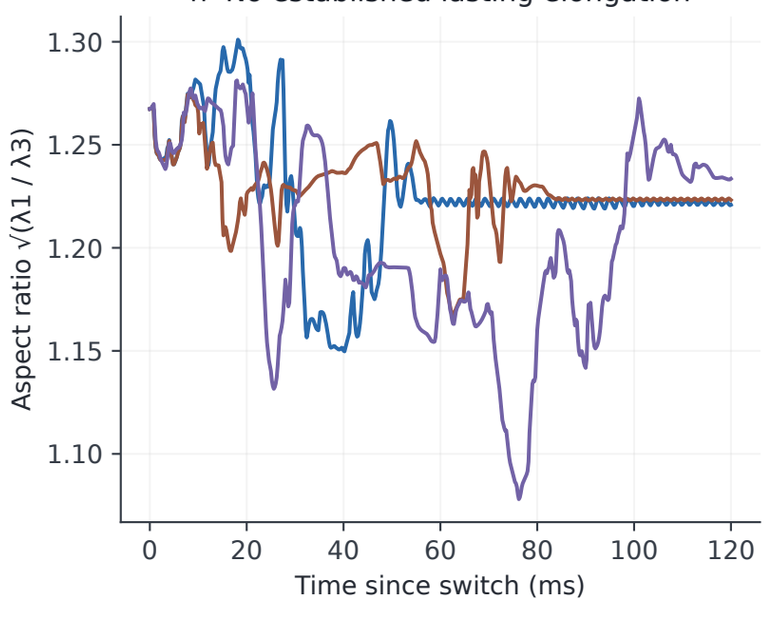
f Eigenvalue contrast above all envelopes



g Switch at mature absolute time 102.2 ms



h No established lasting elongation



i Interpretation

Longest eigenvalue windows above envelopes:
phase 2.6 ms; amplitude 8.8 ms
Both shorter than the 10 ms sustained rule.

TRANSIENT_SHAPE_RESPONSE
NO_ROBUST_ENDPOINT_SHAPE_RESPONSE

All 32 members remain connected.
Persistent split / merge: 0 / 0
No membership exchange.

Same-control stable-repeat differences are zero for raw states; SVD residuals are roundoff only.
EXPLORATORY; field and mobility limits remain.